

Aviral Gupta

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TECHNICAL SKILLS

Languages: Java, Python, C/C++, C#, SQL, JavaScript, HTML/CSS, R, TypeScript

Frameworks: React, Node.js, JUnit, Bootstrap, ASP.NET Core, Flask, Numpy, Selenium, PyTorch, Tailwind

Technologies: Git, Postman, MongoDB, Linux, AWS, Azure, Docker, MySQL, SOAP UI

EXPERIENCE

Software QA Analyst | *TransLink*

March 2025 – Current

- Automated **over 50% of 170+** CompassCard website test cases by developing Selenium and Postman scripts, **reducing manual testing time by 70%**.
- Performed load testing simulating 1000+ users using Apache JMeter, leading to a **20% reduction** in server response time
- Migrated the Student UPASS project from on-prem servers to **Azure Landing Zone** (ALZ), saving approximately **\$5,000 per month** in operational costs.
- Led** the UPASS testing team by designing and implementing robust testing infrastructure and processes, ensuring a healthy, maintainable test suite for **scalability**.

Software QA Engineer Intern | *Square One*

Sep 2023 – Dec 2023

- Developed and executed **200+ regression and automated test cases** using Cypress and Postman, eliminating critical bugs in insurance policies and underwriting webpages while **reducing test suite runtime by 2h 30m**.
- Automated mobile testing across iOS and Android by leveraging LambdaTest's cloud-based virtual machine platforms, ensuring app compatibility across multiple OS versions and devices.
- Engineered complex **SQL queries in relational databases** to retrieve and validate customer records, ensuring **100% accuracy** of issued insurance policies and preventing data loss.
- Optimized batch job test matrices during codebase migration, **reducing total execution time by 71%** (from 5 hours to 1h 30m) and streamlining deployment from test to production servers.

Software Developer Intern – Research Assistant | *UBC Sauder*

Mar 2023 – Aug 2023

- Boosted data collection efficiency **by 50%** by developing a wrapper API for Twitter's REST API in Python, automating the scraping of 1,000,000+ tweets on S&P 500 companies.
- Built a **linear regression model with scikit-learn**, leveraging ESG report data to predict carbon emissions for S&P 500 companies over 30 years.
- Implemented cross-validation techniques, improving model's prediction **accuracy to 93.8%**, enhancing performance and improving reliability.
- Cleared 100% of backlog issues in every 2-week sprint cycle in an Agile environment, keeping the codebase bug-free and reducing technical debt.

PROJECTS

GTOForger – AI Poker Solver | *Python, PyTorch, C++, Ncurses*

- Created a game-theory-optimized poker solver by building a C++ terminal UI with the ncurses library, allowing players to input scenarios and simulate outcomes.
- Implemented counterfactual regret minimization (CFR) in Python, cutting decision-making speed by 50% in the game tree and minimizing risk through Nash equilibrium strategies.
- Trained a supervised learning model on 400,000+ poker hands improving bluff-catching accuracy by 80% and enhancing overall strategic performance.

OkTherapy – Personalized Therapy Matching Website | *React, CSS, HTML, JavaScript, MongoDB, ReduxJS, Chakra-UI*

- Built a dynamic web application using ReactJS, enabling users to find therapists tailored to their specific needs.
- Designed UI using Chakra UI for responsive, user-friendly, and visually appealing design components, enhancing the user experience across various devices.
- Implemented the Sign-In/Sign-Up page using Auth0 for secure and encrypted user authentication.
- Developed a state-managed questionnaire using Redux, allowing users to input preferences for therapists.
- Tested and validated API calls with Postman, verifying seamless communication between the frontend and database.

EDUCATION

University of British Columbia

Sep 2020 – Dec 2024

Bachelor of Science in Computer Science

- Dean's List top 15% academic achievers